CSE4062

Introduction to Data Science and Analytics

Spring 2023

FINAL REPORT

IT Support Tickets

MEMBERS

Data Statistics

There are 40,008 rows and 18 columns in the dataset.

Target Attribute: **URGENCY**

#	Column Name	Type	Important
1	PRIORITY	Nominal	Yes
4	URGENCY	Nominal	**Target**
5	IMPACT	Nominal	Yes
7	ASSIGNEE	Text	Yes
8	POZISYON	Nominal	Yes
9	IS_BILGISI	Nominal	Yes
11	Completion Time	Numeric (float)	Yes
18	ISSUE_TYPE	Text	Yes

Feature Selection

There are 18 columns in the dataset including target column *URGENCY*. We picked them 5 columns to use in our training process by using different methods. Before running feature selection algorithms we did one-hot encoding to categorical values of the dataset. Then we determined the columns will be used in the training by using random forests, information gain and manual selection. Here is the list of selected features.

Table 1. Selected features

	Random Forests (I)	Information Gain (II)	Manual Selection (III)
IMPACT_Operation Continue With Workaround	x	x	
IMPACT_Some Processes Stop on Production	x		x

PRIORITY_Major	x	x	x
PRIORITY_Minor		x	
WAITING_HOURS_AT_CUSTOMER	x	x	
Completion Time	x	x	x
ISSUE_TYPE_Incident			x
COMMENT_COUNT			x

Algorithms Tested

- Support Vector Machine
- Stochastic Gradient Descent
- Decision Tree
- Naive Bayes
- Random Forest
- Multi-Layer Perceptron
- DBSCAN
- K-Means
- Agglomerative Clustering
- Apriori

Results

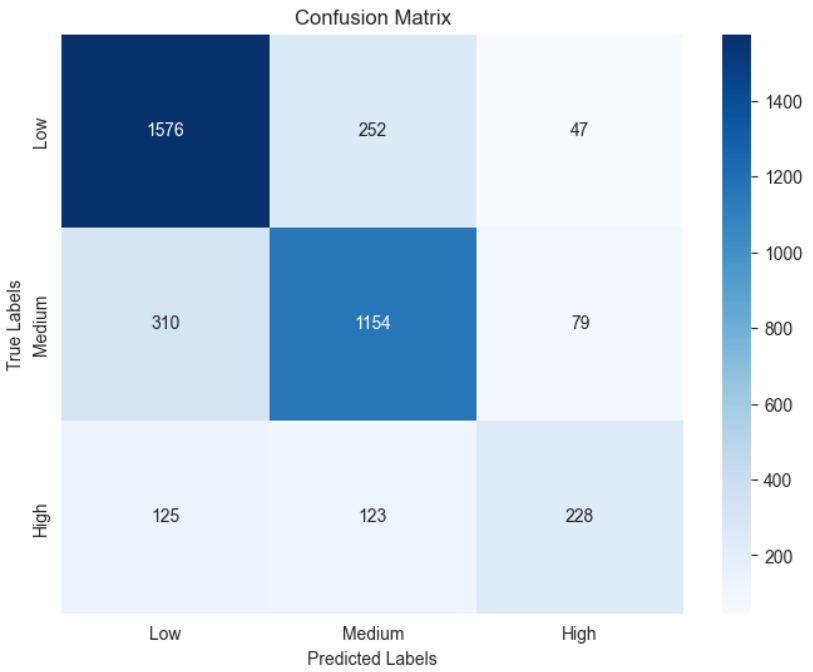
Table 2. Performances of each algorithm tested with three different feature subset sorted by F1 scores

Algorithm	Features	Accuracy	F1	Precision	Recall	F1-Micro	F1-Macro
Random Forest	I	0,76	0,75	0,75	0,76	0,76	0,70
Random Forest	II	0,72	0,71	0,71	0,72	0,71	0,66

Decision Tree	I	0,70	0,70	0,70	0,70	0,71	0,65
Decision Tree	II	0,68	0,67	0,67	0,68	0,68	0,62
MLP	I	0,68	0,67	0,67	0,68	0,70	0,62
Random Forest	III	0,65	0,65	0,64	0,65	0,65	0,57
Decision Tree	III	0,65	0,64	0,64	0,65	0,64	0,57
MLP	III	0,64	0,63	0,64	0,64	0,64	0,53
MLP	II	0,66	0,62	0,65	0,66	0,66	0,59
SGD	III	0,59	0,55	0,52	0,59	0,58	0,41
Naive Bayes	II	0,60	0,54	0,57	0,60	0,60	0,41
SGD	I	0,57	0,54	0,55	0,57	0,54	0,43
SGD	II	0,56	0,52	0,54	0,56	0,59	0,43
Naive Bayes	III	0,51	0,51	0,44	0,46	0,51	0,34
SVM	III	0,53	0,47	0,47	0,53	0,53	0,35
Naive Bayes	I	0,54	0,47	0,51	0,54	0,54	0,37
SVM	I	0,51	0,41	0,58	0,51	0,51	0,30
SVM	II	0,51	0,41	0,58	0,51	0,51	0,30

The best model is random forest with the feature set I. The confusion matrix of it's the best fold can be seen below.

Figure 1. Confusion matrix of the best model



The second best model (with a different algorithm) is decision tree with the feature set I. The t_test scores between them given below.

Table 3. T-test between random forest model and decision tree model

t-statistic	10.597172769757407
p-value	7.638353991567266e-19

Figure 2. All metrics of all models when feature subset I is used

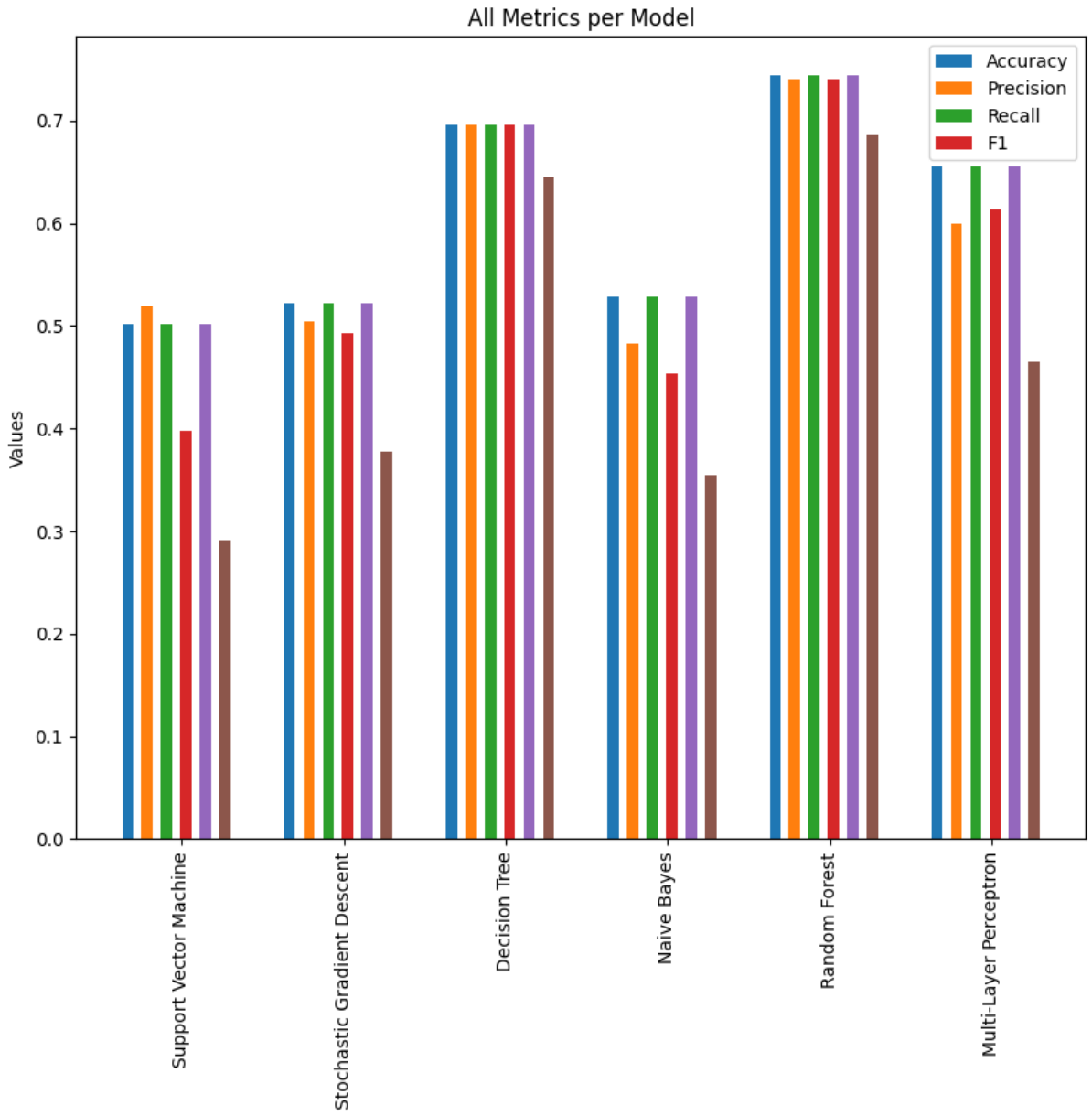
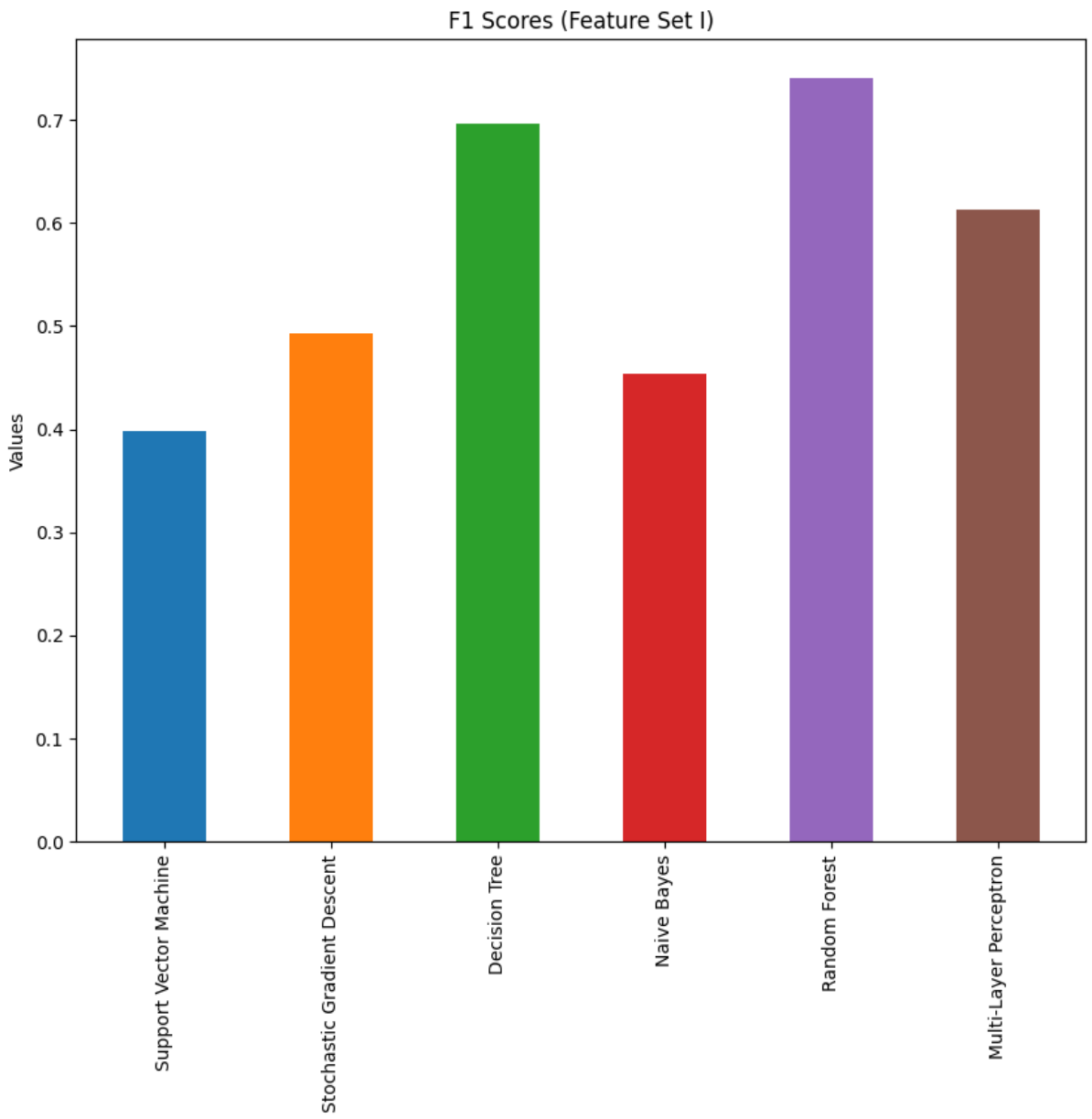


Figure 3. F1 scores of different models when feature subset 1 is used



Apriori Results

Apriori algorithm executed on these columns only.

- IMPACT_Operation Continue With Workaround
- PRIORITY_Major
- IMPACT_Some Processes Stop on Production
- IMPACT_Operation Continue With Workaround
- PRIORITY_Minor

Findings

1. The first association rule shows that when there is an impact on some processes stopping production, there is a 44.0% chance of having a minor priority. This rule has a lift value of 1.153, suggesting a slightly positive correlation between the antecedent and consequent.
2. The second association rule indicates that when the priority is minor, there is a 73.8% chance of impacting some processes to stop production. This rule also has a lift value of 1.153, indicating a slight positive correlation between the antecedent and consequent.
3. The third association rule suggests that when the priority is minor, there is an 18.4% chance of continuing operations with a workaround. This rule has a lift value of 1.112, indicating a slightly positive correlation.

4. The fourth association rule states that when operations continue with a workaround, there is a 42.4% chance of having a minor priority. The lift value of 1.112 suggests a slight positive correlation between the antecedent and consequent.
5. The fifth association rule shows that when operations continue with a workaround, there is a 52.2% chance of having a major priority. The lift value of 1.036 suggests a weak positive correlation between the antecedent and consequent.

Results

	antecedents	**consequents**	**antecedent support**
4	(IMPACT_Some Processes Stop on Production)	(PRIORITY_Minor)	0.640080
5	(PRIORITY_Minor)	(IMPACT_Some Processes Stop on Production)	0.381633
2	(PRIORITY_Minor)	(IMPACT_Operation Continue With Workaround)	0.381633
3	(IMPACT_Operation Continue With Workaround)	(PRIORITY_Minor)	0.165301
0	(IMPACT_Operation Continue With Workaround)	(PRIORITY_Major)	0.165301

1 (IMPACT_Operation Continue With Workaround) 0.503225

	consequent support	support	confidence	lift	leverage	conviction
4	0.381633	0.281728	0.440145	1.153320	0.037452	1.104512
5	0.640080	0.281728	0.738217	1.153320	0.037452	1.374879
2	0.165301	0.070149	0.183814	1.111997	0.007065	1.022682
3	0.381633	0.070149	0.424374	1.111997	0.007065	1.074252
0	0.503225	0.086209	0.521530	1.036375	0.003026	1.038257
1	0.165301	0.086209	0.171313	1.036375	0.003026	1.007256

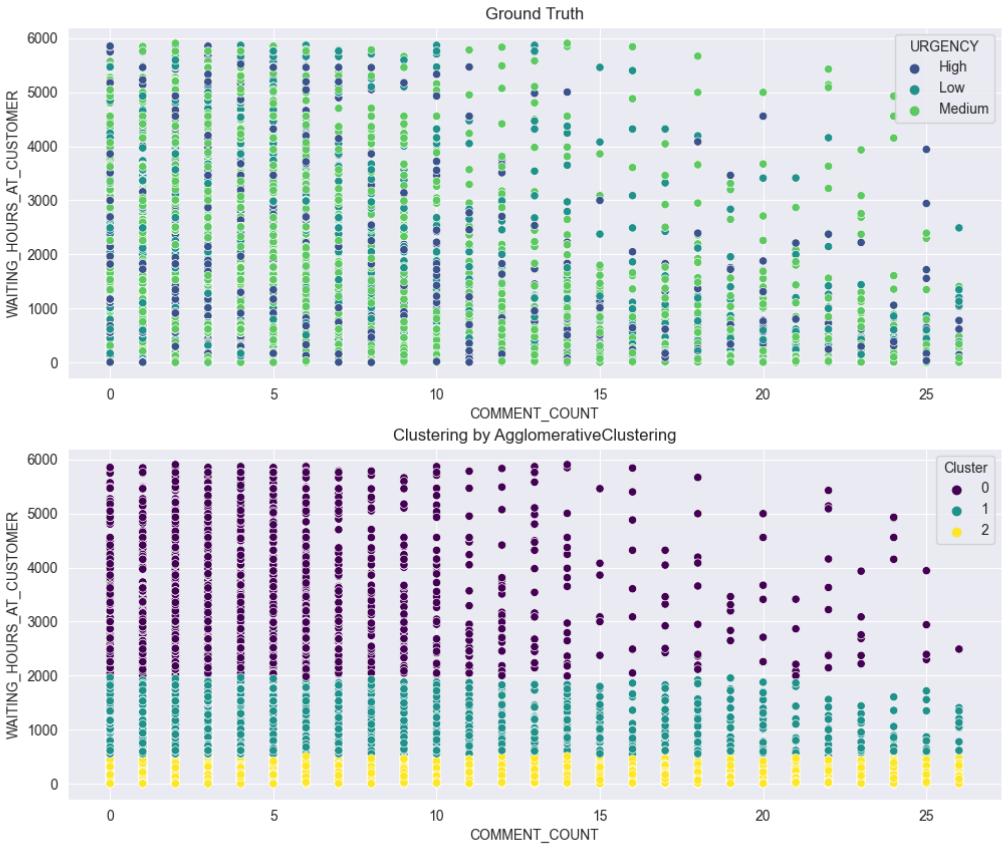
	zhangs_metric
4	0.369354
5	0.214982
2	0.162875
3	0.120662
0	0.042049
1	0.070652

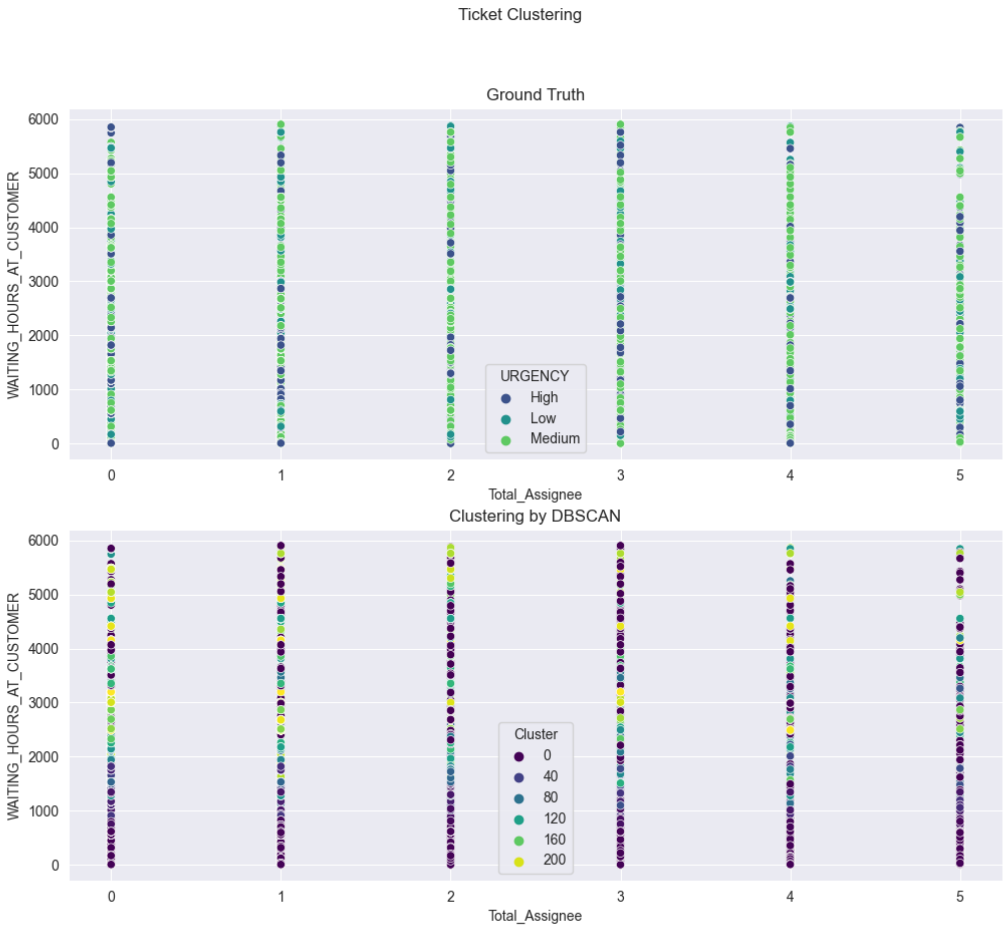
Clustering Results

We created 2 length combinations of all numerical columns listed below and run DBSCAN, K-Means and Agglomerative Clustering algorithms. We couldn't find any interesting pattern by using these algorithms. We obtained 63 different results with graphs. You can see some of them in the appendix section.

1. Complition Time
2. Total_Assignee
3. Total_Worklog_Assginee
4. Total_Log_Hours_Assignee
5. COMMENTOR_COUNT
6. COMMENT_COUNT
7. WAITING_HOURS_AT_CUSTOMER

Ticket Clustering




Ticket Clustering
Ground Truth
WAITING_HOURS_AT_CUSTOMER
6000
5000
4000
3000
2000
1000
0
URGENCY
High
Low
Medium
0
1
2
3
4
5
Total_Assignee
Clustering by DBSCAN
WAITING_HOURS_AT_CUSTOMER
Cluster
0
40
80
120
160
200
Total_Assignee



Conclusion

Random Forest classification algorithm can be used in real life after doing some fine-tuning in our opinion. The error rate can be tolerated in real life.

The Apriori results indicate several associations between different variables. Specifically, there is a correlation between an impact on some processes stopping production and having a minor priority, as well as between a minor priority and impacting processes. Additionally, there is a connection between continuing operations with a workaround and having a minor or major priority. These findings provide insights into the relationships and dependencies among these variables.